

WEEK - 9

Writing Results Section of an Academic Report

Video 1: Structure of the Results Section

Function of the Result Section

The result section should :

- Present the findings of the study
- Report the findings with a less subjective interpretation or no interpretation
- Show your contribution to the field

In the result section

- Findings must be reported and commented on
- Findings can be reported with the help of tables, graphs, or any other visual forms

Presentation and organization of the result section are crucial to gain the attention of the academic reader and community.

Structure of the Result Section

Different ways of organizing a result section

- In the order of research questions or hypothesis
- In themes and sub-themes
- Overall results can be presented in the beginning and followed by the specific results
- Arrange your results in a logical order
- The structure of the result section should contribute to the overall objective of the study

Video subtitle:

“There are different ways of organizing a result section, obviously, what academic report you're writing, what research you have been into. Whether you are doing a sample survey or your research is heavily based on lab experiment and the findings of the lab experiment, or you were doing a social science or humanities research where you engage with lot of texts and thematic and argument. Depending on that, your results section would vary.”

Video 2: Different Components of the Results Section

How to Write the Result Section?

- Provide background information and briefly state the finding
- Present the data in visual form
- Elaborate the data
- Provide an explanation and supporting details
- If necessary, Comment on the data
- Explain the difference in results if it doesn't support your hypothesis
- Report anomaly in data

Results Section: Example 1 (Type 1)

To evaluate the effects of missing data and imputation strategies on model fairness and performance, we explore the experimental data aligned to the three main factors of the experimental design, i.e., i) percentage of missing data, ii) machine learning model, and iii) imputation strategy and their effect(s) on fairness and performance metrics.

(stating the structure and providing background for the result section.)

Source: Cropper, A., & Dumandic, S. (2022). Inductive logic programming at 30: A new introduction. *Journal of Artificial Intelligence Research*, 74, 765-850.

Results Section: Example 2 (Type 2)

Clearly, a strongly popular partition is also popular, and a popular a partition, interpreted as a probability distribution with singleton support, is mixed popular. Furthermore, every coalition structure supporting a mixed popular partition is Pareto optimal. This already follows from a more general statement by Fishburn (1984, Prop.3). We give a simple proof for completeness.

Proposition 2: Let $p = \{(11, p_1), \dots, (1k, p_k)\}$ be a mixed popular partition. Then, for every $i \in [k]$ with $p_i > 0$, TT_i is Pareto optimal.

(Presenting the result with proposition.)

Source: Cropper, A., & Dumanic, S. (2022). Inductive logic programming at 30: A new introduction. *Journal of Artificial Intelligence Research*, 74, 765-850.

Results Section: Example 3 (Type 3)

There is a also a notable increase in outliers. Whilst neither of these observations is surprising, that different imputation strategies appear to affect specific performance and fairness metrics differently is. This would initially indicate that care is needed when handling missing values, and that significant testing is needed when treating missing data.

(Comment or interpretation of the data)

Source: Cropper, A., & Dumanic, S. (2022). Inductive logic programming at 30: A new introduction. *Journal of Artificial Intelligence Research*, 74, 765-850.

Results Section: Example 4 (Type 4)

Overall, considering the first part of the research question posed (whether the imputation strategy affects the fairness and/or performance of a machine learning model), an initial inspection of the distribution of performance and fairness outcomes visualized in Figures and 2 suggests that the choice of imputation strategy significantly affects both the performance and the fairness of the resulting predictions.

(Connecting your research question to findings.)

Source: Cropper, A., & Dumanic, S. (2022). Inductive logic programming at 30: A new introduction. *Journal of Artificial Intelligence Research*, 74, 765-850.

Video 3: Using Figures in the Results Section

What Is the Use of Figures or Tables?

- Effectively present the data
- Logically organize the data
- Present the result clearly and concisely
- Effectively communicate the results to the academic community
- Facilitate the result comprehension
- Present the complex process precisely

Two Ways to Explain the Figures or Tables

- Reporting the figures or tables
 - To describe figures or tables but not to comment on the data
 - Used when the author wants the reader to make the inference
 - Example: Figure 2.1 explains the correlation between smoking and hypertension in adult females
- Interpreting the figures or tables
 - To describe and interpret the figures or tables
 - Used when the author wants to state the interpretation explicitly
 - Example: Smoking increases the risk of hypertension in adult females (Table 2.2).

Video subtitle:

“In the first instance, you are just saying what the figure is explaining, and what the figure is showing. Then the data is coming in a form of a table format just to support your claim and show this is the claim. Whereas in the second example, you are also not just presenting the figure or the table, but you're also reading them. It was just saying this is a way like I'm reading, I'm interpreting, and then you're providing your table so your source from where you are reading certain results.”

Figures and Tables: Example

Envy-fearness	Objective function	Lower bound	Upper bound
market envy-fearness	Social welfare	$m - \epsilon, \forall \epsilon > 0$ (Claim 20)	m (Claim 19)
	Revenue		
agent envy-fearness	Social welfare	$(2 - \epsilon), \forall \epsilon > 0$ (Claim 25)	2 (Theorem 24)
	Revenue		

Table 3

(Reporting the figures or tables)

Table 3: How preselection can improve the objective functions of envy-free solutions in multi-unit markets with identical items. All lower bounds hold, even in the case of single-minded buyers.

(Interpreting the figures or tables)

We show how preselection can improve the revenue and the social welfare of both market and agent envy-free solutions. These results are summarized in Table 3.

Video 4: Language Use in the Results Section

Tenses Used in the Result Section

- Past tense is predominantly used to report the data
 - Example: In brief, more than 60% of students born after 2000 faced mental stress in their teens.
- Present tense is used to report the table and interpret the data.
 - Example: Table 2.2 illustrates the income difference between students from international schools and normal private schools.
 - Example: The findings show parents in rural India want their kids to be educated in English medium schools.

Impersonal or Personal Style?

Impersonal Style

- Most authors use an impersonal style to present their data. It helps achieve objectivity in their writing.
- Example: This phenomenon reflects the need for a robust framework to develop biliteracy among school children.

Video subtitle:

“Generally, in science research, authors use impersonal style to present the data and generally most authors do that. Impersonal style helps in achieving objectivity in our writing, so that it's not about me and we and our opinion and our reading, it's a truth which is there, which came from certain experimental results and impersonal style helped to build that impression of, this is something which is presented as a critically objective manner when the person himself or herself is not important, but the result and the dataset and the findings are much more important, so that's why the impersonal style is preferred in scientific research particularly.”

Personal Style

- Experts in the field or established authors might use a personal style to report or interpret the results.
- Example: We report that a robust framework is necessary to develop biliteracy among school children.

Video subtitle:

“Experts in the field, or established authors, might use the personal style sometimes to report or interpret their result. Sometimes when you are arguing certain points, or your opinion is important rather than so much of the facts and your findings, then you might use a personal style, but generally, in scientific research, the preferable form is impersonal style.”

Avoid Adding Value to the Data

- Do not add value or adjectives to your data.
- Example: Most interestingly, the tables show that more Artificial Intelligence (AI) projects are approved by the government after the recent International summit on AI.
- Reporting should make it an interesting observation.
- Example: The tables show that the government has approved 580 Artificial Intelligence projects in the last three months, between April to June. There is a 30% increase in the government's approval rate after the recent International summit on AI.

Video subtitle:

"Instead, your reporting should make the reader feel that it is an interesting observation rather than you are saying 'most interestingly' or something like that, so it's preferable you don't use those values to your data."

Writing Discussion Section of an Academic Report

Video 1: Structure of the Discussion Section

Discussion Section of an Academic Report

- Appears toward the end of a report and precedes the conclusion
- Provides analysis and interpretation of the findings of the research
- Focuses on offering an explanation and evaluation of what has been found in the research
- Explains how the research findings
 - Provide solutions to the research question
 - Offer justification for the methods applied to conduct the research
- Analyzes the findings in the context of the literature review
- Provides support to the argument presented in the research

Purpose of Discussion Section

- Presents description, interpretation, and analysis of the research results or findings
- The key components of the discussion are
 - An analytical review of the findings
 - Insights on its relevance in the context of the entire research
 - Discussion on all the preceding sections of a research report
 - Insights into the logical connection that exists between the sections
- Discussion section
 - Provides an analysis of the results
 - Explains why the results are significant in the respective field of research

Structure of Discussion Section

- Introductory paragraph
 - Outlines the results of the research
 - Provides an idea on the supporting evidence
 - Explains how the evidence are related to the results
- Intermediate paragraph
 - Comprises the body of the discussion
 - Offers a comprehensive interpretation of the key findings
 - Provides comments on the shortcomings of the research
 - Analyzes any unexpected results
- Concluding paragraph
 - Summarizes the discussion
 - Offers an understanding of the connection between the research hypothesis and the findings
 - Highlights the rationale of the discussion section
 - Mentions the scope for future research

Video 2: Different Components of the Discussion Section

Components of Discussion Section

- **Summary of key findings**
 - Begins with a summary, including
 - The key findings
 - An analysis
 - Includes how these are related to the research questions and research hypothesis
- **Context**
 - The research findings must be analyzed in the context of the literature reviewed
 - This analysis reflects the significance of the research in the respective field of study
- **Interpretation of findings**
 - The researcher
 - Discusses the nature of the findings
 - Comments on whether the results have been per expectations
 - Highlights any unexpected results
- **Discussion on unexpected results**
 - The unexpected results must be discussed, and an interpretation must be provided on
 - Why the results are unexpected
 - How or whether those results impact the research
 - **Video subtitle:**
 - “The unexpected results have to be discussed and an interpretation has to be provided on the results are unexpected and how or whether those results impact the research. The point is, you should be very clear and transparent to readers and certain things which are not like moving as per the plan or the results which are not everything or everything is coming as per the expectation, that doesn't mean you will not provide the detail of those results, so it's very important to provide a detail of even the unexpected results.”
- **Implications of the findings**
 - Discusses the relevance of the findings in the context of the overall research
 - Explores the new understandings, theoretical or practical, of the research contribution
- **Limitations of the research**
 - Highlights the limitations of the study
 - Acknowledging that the limitations add clarity to the findings
 - Increases the credibility of the researcher
- **Scope for future research**
 - Includes an outline of the scope for future research
 - Must include a recommendation about the potential areas of research
- **Conclusion and restatement of findings**
 - Concludes with a restatement of the findings and their implications

Discussion Section and Its Significance

Results Section	Provides the findings and the rationale behind the results	Conclusion	Discussion Section
Discussion Section	Offers a detailed interpretation and evaluation of those findings	Presents a summary of the overall research report Briefly and generically restates the key points of the overall report	Offers a comprehensive and rigorous analysis of the research results Includes the implications and limitations of the study

The discussion section is the last segment **before** the conclusion that provides an in-depth research analysis.

Video 3: Writing About the Limitations of the Study

Limitations of a Study

- A common factor associated with research
- Every research has some shortcomings or weaknesses
- It is essential to highlight it because:
 - Adds more authenticity to the research
 - Increases the credibility of the researcher
 - Reflects that the researcher has a proper understanding of the study topic
 - May impact the research conclusion
 - Its analysis creates scope for further research
- **Video subtitle:**
 - “The idea is rather than the reader of your research or a report, like finding out, okay, this is where the authenticity, like lax or something like that and pointing out with your limitation, it's better the researcher himself or herself is stating this is where the research has certain limitation. Also, it increases the credibility of the researcher that you were like as a researcher, finding out the limitation in research and you're also stating in the discussion section. It reflects the researcher has proper understanding of the topic of the study. The part you could do and the part you couldn't cover because of several reasons but you are announcing or stating it in the discussion section. The limitations may impact the conclusion drawn in a research and analysis of the limitation creates scope for further research. Very important. You are showing there could be a gap or limitations for the future researcher could take up that kind of a gap and then the research could proceed in a kind of a further and furthermore. The limitation of study are also included towards the end of our discussion section of research report, this is generally the area or a part or a section where you include our research.”

Limitations That Commonly Impact a Study

- Time constraint
 - Constraint of time and inability to meet deadlines
- Inaccessibility of information or data
 - Limited accessibility or inaccessibility of information or data
 - Unavailability of relevant literature
- Limitations in research design
 - The nature of research questions
 - The hypotheses and research methods to be adopted
 - A shortcoming in the research design
- Limitations in research methodology
 - Lack of insight regarding adopting an effective research methodology
- Biased viewpoints
 - Biased ideas and perspectives regarding the selection of research tools and literature

Writing Limitations of a Study

Components in writing the limitations of a study

- Description of limitations
 - Clearly and concisely describes each limitation of the research
 - Reflects the researcher's analytical skills
- Explanation
 - Examines and explains the reasons behind the limitations
- Impact of limitations
 - Assesses and analyzes the impact on the overall research
- Scope for further research
 - Includes discussion on how the limitations can create scope for further research

Video 4: Language Use in the Discussion Section

Writing the Discussion Section

The Discussion section of a research report is

- A crucial and challenging part to write
- Provides description and analysis of research findings
- Must be written concisely and comprehensibly, avoiding jargon of the field
- Subsections and subheadings add clarity to the analysis
- Tables or figures must be numbered and titled
- The significant portions of the research findings must be highlighted
- A researcher must be clear and specific about stating the limitations of the research

Language Use in Discussion Section (How to start (phrases used in) the Discussion Section)

- Research findings (How to start (phrases) Research Findings)
 - "This study suggests that"
 - "The key findings that emerge from the present study"
 - "The findings demonstrate an understanding of"
 - "This finding is better than the results that are currently accepted"
- Summary of the results
 - "This section summarizes the findings in a concise fashion"
 - "The purpose of the study was to provide a better insight on"
 - "The findings of the present research support the hypothesis of"
 - "The findings of the present study offers evidence to support the research question"
- Comparison with prior research
 - "In accordance with the previous research"
 - "The findings of the present study are consistent with"
 - "Contrary to the results of...it is to be determined"
 - "Whereas past studies have demonstrated... (citation)..the present research claims"
- Research Results
 - "The results of the present research focus on"
 - "The findings strongly indicate that"
 - "The interpretation of the findings suggest"
 - "Finally, we found evidence of"
- Research limitations
 - "There are two potential limitations of the research - (specific mention of those limitations)"
 - "Certain limitations of the present research can be addressed in future research"
 - "An apparent limitation of the methodology is"
 - "A major limitation has arisen due to the lack of"
- Scope for future research
 - "The present research establishes that much work has to be done for a complete understanding of the topic"
 - "In addition to those already stated, some other potential avenues of research are"
 - "In terms of future study, it would be useful to examine the current findings"
- Closing statements
 - "The present research contributes to a growing body of evidence"
 - "To summarize, the present study establishes"
 - "The present research has enhanced the understanding of the (research topic)"

Writing Conclusion Section of an Academic Report

Video 1: Structure of the Conclusion Section

Conclusion Section

- Usually shorter than introductions
- Can be of several types
- Summarizing the main ideas
- Confirming answers to research questions

Conclusion Section: Structure

- Achievement of research objectives
- Answers to research questions
- Implications of research findings on real-world applications
- Summary of the main ideas
- Limitations of the study (if applicable)
- Comparison with results of other studies (if applicable)

Conclusion Section: Example

“We have introduced a family of novel tree-regularization techniques that encourages the complex decision boundaries of any differentiable model to be well-approximated by human-simulable functions, allowing domain experts to quickly understand and approximately compute what the model is doing. Overall, our training procedure is robust and efficient.”

“More broadly, our general training procedure could apply tree regularization or other procedure-regularization to a wide class of popular models, helping us move beyond sparsity toward models humans can easily simulate and thus trust.”

Video subtitle:

“This is just an idea, this is an instance of how you can design your conclusion. But these are the key points to keep in your mind when you structure your conclusion section. Please remember this is a very important part of your academic report. You should be very careful when you design the conclusion part.”

Video 2: Components of the Conclusion Section

Conclusion Section: Components

- Summarizes the main ideas or arguments in one or two sentences
- Mentions how this study contributes to or improves on existing research
- Shows that the research objective has been achieved or research questions have been answered
- Suggests directions for future research

Conclusion Section: Example

- “This paper presented a novel semi-exact method for the maximum weight clique problem (MWCP), which aims to solve large graphs within a short time limit. The method interleaves clique finding and graph reduction.”
 - In the first couple of sentences...
 - The authors concisely presents the method introduced in the paper
- “Experiments on large real-world graphs show that FastWClq finds better solutions than state-of-the-art algorithms while using less time on most instances.”
 - In the next statement..
 - The authors mention how this method is promising and how it benefits the field
- “As shown in this work, semi-exact algorithms take the advantage of solving and proving, and seem a promising direction for solving large combinatorial optimization problems.”
 - In the fourth (Third point above) statement
 - The authors mention how the suggested method satisfies the research objective
- “A significant direction for future work is to apply the semi-exact method to other combinatorial optimization problems.”
 - In the last statement
 - The authors suggest a future research path

Video 3: Giving Directions for Further Research

Further Research

- Research addresses one possible research gap.
- It adds new knowledge to the existing body of knowledge and thereby raises new questions.
- Conducting research in a particular area allows you to open new avenues of research in the same area.

Giving Directions for Further Research

- Include some suggestions for the possible research areas in the conclusion section
- Seek to address the limitations in future studies
- Describe the new phenomenon and mention the need to explore it further
- The scope and scale of the study can be increased in subsequent years based on:
 - Availability of funding
 - New technology
 - Other factors
- Compare your study with others and find ways of improving the methods or techniques used

Research Direction: Examples

- We intend to consider a more complex case study and to extend our work with a number of additional aspects.
- Incorporating a degree of norm violation, and altering the sanctions used to enforce the norms, is another interesting extension of this work.
- Similarly, the study here reported focused on a single MAS objective and on a boolean evaluation of such an objective. In future work, we also intend to extend the proposal to multiple labels, thereby supporting multiple objectives.

Source: Dell'Anna, D., Alechina, N., Dalpiaz, F., Dastani, M., Logan, B. (2022). Data Driven Revision of Conditional Norms in Multi-Agent Systems. Journal of Artificial Intelligence Research. 75. 1549-1593.

Video 4: Language Use in the Conclusion Section

Language Use in the Conclusion Section

- Present the key points concisely
- Give a strong take-home message
- No new arguments are introduced
- Raise questions or set the context for future research
- Present next steps to be taken

Phrases to be Used for Limitation

- The scope of this study was limited to...
- However, this study does not consider..
- Owing to a small sample size, the study may not represent...
- The findings of this study are only relevant in terms of...
- The data suggests that we still require..
- It is difficult to arrive at conclusion/a generalization with respect to...

Phrases to be Used for Further Research

- In future work, the following need to be considered...
- Investigating in future studies is a possible avenue to be explored.
- A key issue to be considered in future studies is...
- With advancements to technology in the future, we may be able to...
- This phenomenon requires additional exploration as...
- These initial findings can be confirmed further by...
- Earlier studies in the domain can be replicated by...